```
100 'MEAN AND STANDARD DEVIATION USING RUNNING STATISTICS
110 '
120 DIM X[511]                    'The signal is held in X[0] to X[511]
130 '
140 GOSUB XXXX                    'Mythical subroutine that loads the signal into X[ ]
150 '
160 N% = 0                        'Zero the three running parameters
170 SUM = 0
180 SUMSQUARES = 0
190 '
200 FOR I% = 0 TO 511             'Loop through each sample in the signal
210   '
220   N% = N%+1                   'Update the three parameters
230   SUM = SUM + X(I%)
240   SUMSQUARES = SUMSQUARES + X(I%)^2
250   '
260   MEAN = SUM/N%    'Calculate mean and standard deviation via Eq. 2-3
270   VARIANCE = (SUMSQUARES -  SUM^2/N%) / (N%-1)
280   SD = SQR(VARIANCE)
290   '
300   PRINT  MEAN   SD   'Print the running mean and standard deviation
310   '
320 NEXT I%
330 '
340 END
```

TABLE 2-2

Before ending this discussion on the mean and standard deviation, two other terms need to be mentioned. In some situations, the *mean* describes what is being measured, while the *standard deviation* represents noise and other interference. In these cases, the standard deviation is not important in itself, but only in *comparison* to the mean. This gives rise to the term: **signal-to-noise ratio (SNR)**, which is equal to the mean divided by the standard deviation. Another term is also used, the **coefficient of variation (CV)**. This is defined as the standard deviation divided by the mean, multiplied by 100 percent. For example, a signal (or other group of measure values) with a CV of 2%, has an SNR of 50. Better data means a *higher* value for the SNR and a *lower* value for the CV.

Signal vs. Underlying Process

Statistics is the science of interpreting *numerical data*, such as acquired signals. In comparison, **probability** is used in DSP to understand the *processes* that generate signals. Although they are closely related, the distinction between the **acquired signal** and the **underlying process** is key to many DSP techniques.

For example, imagine creating a 1000 point signal by flipping a coin 1000 times. If the coin flip is heads, the corresponding sample is made a value of one. On tails, the sample is set to zero. The *process* that created this signal has a mean of exactly 0.5, determined by the relative probability of each possible outcome: 50% heads, 50% tails. However, it is unlikely that the actual 1000 point signal will have a mean of exactly 0.5. Random chance